



MINISTRY OF EDUCATION AND TRAINING

FPT UNIVERSITY

Capstone Project Document

Project Title

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Despite our best efforts to make this thesis as complete as possible, shortcomings and limitations are inevitable. We sincerely hope to receive your understanding and constructive feedback from our supervisors and readers.

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Abstract

A good abstract is a stand-alone summary of the paper, and should summarize the key components of the manuscript. Generally, the abstract should be concise and informative within 200-300 words. As an abstract is a separate section, it should be a self-containing text (no abbreviations, no references, no URLs, no undefined concepts, etc.). For research articles, abstracts should give a pertinent overview of the work. We strongly encourage authors to use the following style of structured abstracts, but without headings: (1) Background: Place the question addressed in a broad context (research background) and highlight the purpose of the study; (2) Methods: Describe briefly the main methods or treatments applied; (3) Results: Summarize the article's main findings; and (4) Conclusions: Indicate the main conclusions or interpretations.

Keywords: cybersecurity, information security, threat analysis, resilience

List of Abbreviations

Acronym	Definition
DSP	Digital Service Providers
EU	European Union
GSP	Global Stability Protection
OES	Operators of Essential Services

Chapter 1

Introduction

1.1 Background

The explosive growth of the Internet of Things (IoT) has led to the connection of billions of devices, continuously generating massive amounts of data. However, data processing at the edge (Edge Computing) faces significant challenges due to hardware limitations [1]. A specific characteristic of most IoT devices is their severe constraints in computational capabilities, storage space, and battery power [2]. In this context, Intrusion Detection Systems (IDS) play a pivotal role in monitoring traffic and protecting networks from cybersecurity risks, especially complex attacks. Recently, Federated Learning (FL) has emerged as an ideal decentralized solution. This approach allows multiple devices to collaboratively train a machine learning model without sharing raw data, thereby resolving data privacy issues and minimizing communication costs.

1.2 Problem Statement

Despite providing an effective distributed solution, deploying IDS in real-world IoT environments via Federated Learning still encounters core challenges. The first is the risk of privacy leakage and the lack of secure encryption mechanisms during transmission: During the transmission of weights or gradients between devices, users' sensitive information remains at risk of leakage [3], enabling attackers to reconstruct the original data [4] and execute Model Inversion Attacks [5]. The second is the Data Imbalance problem: In IoT network traffic, normal data accounts for an overwhelming proportion, while attack traffic samples are highly rare. Practical evidence from the UNSW-NB15 dataset shows that there are up to 1,064,987 normal samples but only 22,215 attack samples [6]. This severe imbalance degrades IDS

accuracy [7] and directly hinders the feature extraction performance of deep learning models [8]. The third is the limitation regarding the computational costs of current data generation solutions. Although data generation techniques like Generative Adversarial Networks (GANs), typically WGAN-GP, have been proven capable of generating high-quality synthetic data [9], they require excessively heavy and complex computational costs. This creates a significant barrier when deploying on IoT devices that are already constrained in resources.

1.3 Research Objectives

To address the aforementioned limitations, this research proposes a comprehensive framework with the following specific objectives: Deploy two independent architectural pipelines in parallel within a Federated Learning environment (CVAE + FL + Differential Privacy and CVAE + FL + Secure Aggregation) for comparison and contrast, aiming to find the perfect balance between data quality, privacy, and hardware resource optimization. Integrate advanced secure computation methods, namely Secure Aggregation or Differential Privacy, into the weight transmission process to overcome the lack of secure transmission mechanisms in previous studies and defend against data leakage attacks. The FL model will integrate the CVAE (Conditional Variational Autoencoder) model to replace computationally heavy algorithms like WGAN-GP, aiming to generate synthetic attack data to help balance the dataset for the minority class, optimized for tabular data, thereby enhancing the performance of the IDS.

1.4 Significance of the Study

This research holds significant importance in simultaneously addressing two difficult problems in IoT network security: data quality and privacy. By integrating data generation to balance the minority class using a cost-optimized algorithm and applying a distributed training model accompanied by strict security mechanism options (Differential Privacy or Secure Aggregation), this project provides a comprehensive IDS solution that completely protects sensitive information before data is aggregated.

1.5 Scope and Limitations

- **Scope:** The project focuses on processing within an IDS in an IoT environment, specifically limiting the in-depth scope to attack vectors over the MQTT protocol (DoS/DDoS). The generation of balanced data is performed through the CVAE model combined with a distributed Federated Learning architecture. The experimental data uses the CICIoT2023 dataset combined with an enterprise-simulated IoT network dataset collected and refined by the team.
- **Limitations:** Secure computation and synthetic data generation mechanisms require a certain amount of resources; therefore, they may face challenges when deployed on resource-constrained IoT devices [2].

1.6 Thesis Structure

- **Chapter 1: Introduction:** Presents the project's background, objectives, research scope, and the structure of the thesis.
- **Chapter 2: Literature Review:** Evaluates existing studies and demonstrates how this thesis fits into the broader academic context.
- **Chapter 3: Methodology:** Details the research design, data collection methods, and modeling techniques.
- **Chapter 4: Experimental and Results:** Provides a concise and precise description of the experimental results and system evaluations.
- **Chapter 5: Discussion:** Discusses and interprets the significance of the findings from the experimental results in a practical context.
- **Chapter 6: Conclusion and Future Work:** Summarizes the main contributions and proposes future research directions.

Chapter 2

Literature Review

Reviews relevant existing research and demonstrates how your thesis fits into the broader academic conversation.

2.1 Review of Previous Studies

2.2 Summary of the Literature Review

2.3 Contribution of Research

Chapter 3

Methodology

Describes the research design, data collection methods, and analysis techniques.

- 3.1 Research Design**
- 3.2 Data Collection Methods**
- 3.3 Sampling Data Analysis Techniques**
- 3.4 Limitations of the Methodology**

Chapter 4

Experimental and Results

Provide a concise and precise description of the experimental results, their interpretation as well as the experimental conclusions that can be drawn. You should discuss the results and how they can be interpreted in perspective of previous studies and of the working hypotheses.

4.1 Introduction

4.2 Presentation of Data

4.3 Analysis of Results

4.4 Interpretation of Results

4.5 Comparison with Literature

4.6 Implications of the Results

Chapter 5

Discussion

The findings and their implications should be discussed in the broadest context possible and limitations of the work highlighted.

5.1 Restate the Research Problem or Objectives

5.2 Summarize Key Findings

Chapter 6

Conclusion and Future Work

The 'conclusions' are a key component of the paper. It should complement the 'abstract' and is normally used by experts to value the paper's engineering content. A conclusion is not merely a summary of the main topics covered or a re-statement of your research problem, but a synthesis of key points and, if applicable, where you recommend new areas for future research.

6.1 Conclusion

6.2 Future Work

Appendix A

If Any

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